Nicolas Bonasoro

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Simulation | Space Robotics | Spacecraft Design | Systems Engineering

Education

Embry-Riddle Aeronautical University Daytona Beach, FL
Bachelor of Science in Aerospace Engineering (Dean's List) *December 2026*

Rockland Community College Suffern, NY
Associate of Applied Science in Business Administration *May 2021*

Course Project

South Pole Lunar Relay Satellite and Ground Station | *Senior Design, ERAU* Jan 2026 – Present

- Led systems engineering for a lunar South Pole relay satellite and ground station to support lunar surface missions
- Developed S-band/X-band link budget analysis and supported orbital analysis, ADCS design, CATIA satellite bus design, thermal testing, launch vehicle selection, and Interface Control Document (ICD) development

Research Experience

Robust Autonomous Lunar Landing | *Space Robotics & Generative Estimation Lab, ERAU* Oct 2025 – Present

- Developed and analyzed Chrono::DEM and Chrono::CRM lander simulations to evaluate touchdown behavior, sinkage, slippage, and tip-over risk
- Generated active support-polygon margin heat maps for 180 LHS-1 touchdown simulations and confirmed stability outcomes using footpad-contact classification data; found all 30° cases stable at 1270 kg/m^3 but unstable at 1580 kg/m^3 and 1950 kg/m^3

Lunar Trafficability NASA FSGC | *Space Robotics & Generative Estimation Lab, ERAU* Sep 2025 – Present

- Ran Chrono::DEM and Chrono::CRM simulation cases for lunar cone penetration testing, processed force-depth CSV data, and generated plots to support simulation verification and lunar trafficability analysis
- Coordinated lunar trafficability research workflow through Jira task tracking, researcher mentoring, simulation data analysis, and support for rover-mounted payload assembly

Mars & Moon Landing Site Estimation | *Space Robotics & Generative Estimation Lab, ERAU* Feb 2025 – Apr 2025

- Developed Python workflows to extract 3D Mars terrain meshes from NASA MRO CTX DEM data and generate virtual environments for rover simulation
- Simulated autonomous rover missions with ROS 2 and NVIDIA Omniverse to evaluate path-planning behavior and landing-site hazard constraints

Academic Papers & Presentations

- **N. Bonasoro**, J. Cortese, and C. Kilic, "Terramechanics-Based Comparison of DEM and CRM for Lunar Touchdown and Cone Penetration Applications," Manuscript submission with presentation, AIAA SciTech (2027)
- J. Cortese, **N. Bonasoro**, J. Graff, D. Zhang, M. Salih, G. Ellithy, and C. Kilic, "Closing the Loop on In-Situ Lunar Trafficability," Manuscript submission with presentation, AIAA SciTech (2027)
- R. Nagarajan, **N. Bonasoro**, J. Cortese, C. Kilic, and D. Canales Garcia, "AI-Enabled Physics-Based Framework for Tip-Over Risk Prediction in Lunar Lander Touchdown," paper submission and oral presentation accepted for the 77th International Astronautical Congress (2026)
- R. Nagarajan, **N. Bonasoro**, C. Kilic, and D. Canales Garcia "Geometric Modeling of Multi-Contact Touchdown Dynamics and Stability of Lunar Landers," paper submission and oral presentation accepted for the 77th International Astronautical Congress (2026)
- **N. Bonasoro**, J. Cortese, and C. Kilic, "Terramechanics-Based Comparison of DEM and CRM for Lunar Surface Application," Poster Presentation, Embry-Riddle Aeronautical University's Discovery Day (2026)

Skills

Languages: C++, CUDA, Python, MATLAB, LaTeX
Data Analysis: matplotlib, CSV data processing, batch simulation analysis, plot generation, report generation
Tools: Linux (Ubuntu), Docker, CMake, Git, GitHub, Shell, Visual Studio, ParaView, Blender, Simulink, MS Office, Jira, Overleaf
Simulation: CATIA V5, Unreal Engine, ROS 2, NVIDIA Omniverse, Femap, ANSYS Fluent, Project Chrono (Chrono::DEM (Discrete Element Method), Chrono::CRM (SPH-based continuum deformable terrain model))

Portfolio: nicolasbonasoro.github.io/Nicolas-Bonasoro-Website